

Effective viscosity of partially melted ice in the ammonia-water system

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Abstract. The steady-state deformation of partially melted ice in the ammonia-water system was studied by means of a concentric cylinder viscometer in shear stresses, 10 kPa~0.1 MPa, temperatures, 180~210 K and NH_3 contents, 4.0~8.4 %. The flow law found was of a non-Newtonian power-law type; the stress exponent was 4.0 ± 0.1 . The activation energy at constant melt fractions was 33.7 ± 0.8 kJ/mol, which was close to that of viscosity for aqueous ammonia solutions. However, the effective viscosity of partially melted ice estimated at 0.1 MPa was $10^7 \sim 10^{10}$ Pa s, which is about ten orders of magnitudes larger and smaller than that of the ammonia-water mixture in the liquid and solid phases (below a peritectic point, 176 K), respectively.

Introduction

Various geological activities of icy satellite surfaces have been observed in the recent extensive survey by several spacecrafts, especially Voyagers 1 and 2. Their detailed deformations, mechanisms and implications in their thermal evolutions can be only revealed and understood by use of the rheological properties of constituent materials, rocks, ice and other volatiles. One of the most abundant components of icy satellites is water ice, and ammonia is the next possible candidate [Fegley and Prinn, 1989]. Recently, the flow properties of an ammonia-water system were experimentally investigated by Kargel *et al.* [1991] and Durham *et al.* [1993]. Kargel *et al.* used various compositional samples, NH_3 contents of which were varied from 0 to 57 %, and showed that the flow of the ammonia-water system, the melt fraction of which was larger than 75 %, was nearly Newtonian. Durham *et al.* mainly studied sub-solidus mixtures of H_2O and $\text{NH}_3 \cdot 2(\text{H}_2\text{O})$ at a confining pressure of 50 MPa. They found that the rheological laws for the mixture vary systematically from that of pure ice to that of dihydrate, and that pure dihydrate is about 4 orders of magnitudes less viscous than water ice just below the peritectic temperature. However, the rheology of partially melted ice in the ammonia-water system has never been investigated in detail and the onset of partial melting is considered to be very important in the icy satellite rheology because drastic viscosity changes are expected to occur due to interactions or reactions between crystals and liquids.

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For the purpose of clarifying systematically how the viscosity changes with the increasing melt fraction, we made shear creep experiments of slightly melted ammonia-water ice and the results were discussed with reference to the dependence of viscosity on the melt fraction.

Experiments

Shear creep experiments of partially melted ice in the ammonia-water system were carried out by use of a concentric cylinder viscometer (Fig. 1). The copper cylindrical sample holder (diameter 30 mm and height 30 mm) is connected to a cold head of a closed-cycle helium refrigerator, and its temperature is regulated by the refrigerator and electric heater wound on the cold head. The temperature could be changed from 30 K to 230 K with an accuracy of ± 0.1 K. The rotation rod (diameter 10 mm), made of sintered ceramic with low thermal conductivity (1.6 W/Km), immersed in the sample holder, is rotated with a high-torque servomotor (1 Nm), the rotation angle of which can be read and changed with a resolution of 1×10^{-3} degree. The tangential torque acting on the rotation rod is measured with a torque sensor installed in the rod. In the shear creep tests, a given torque was maintained by rotating the rod according to the torque read every one second. To minimize heat flow from the sample, the whole sample system was put in a vacuum chamber kept around 0.1 Pa. The shear tests were conducted at temperatures from 180 K to 210 K and torque from 0.22 Nm to 0.89 Nm.

The ammonia-water solution was prepared by diluting a 29 % ammonia solution with distilled water, where

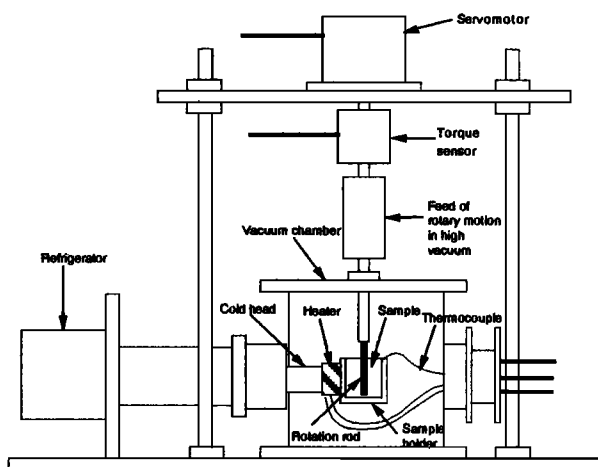


Figure 1. Schematic diagram of the experimental apparatus.

the concentration is a weight fraction of NH_3 in the solution. NH_3 contents studied were 4.0, 6.0 and 8.4 % which correspond to the mole fractions of 4.2, 6.3, and 8.8 respectively. The melt fraction, defined as the mass of liquid phase divided by that of a sample, is calculated by using the phase diagram of $\text{H}_2\text{O}-\text{NH}_3$ compiled by Kargel [1992]. It is a function of NH_3 content and temperature, and its value in our study range is roughly from 12 to 30 %. When the NH_3 solution in the sample holder was cooled below the peritectic point, 176 K, it froze to a complete solid state. Then it was warmed slowly to a test temperature and kept to attain a homogeneous partially melted structure. Slight evaporation from the sample at temperatures lower than 210 K was noted in our experimental procedure usually less than 4 hours, but its effect is negligible to the composition and stress consideration.

When the central rod was rotated without slipping, a concentric shear flow is formed in the sample. The shear stress $\tau_{r\theta}$ at a point (r, θ) in the sample is given by the following equation,

$$\tau_{r\theta} = \frac{M}{2\pi hr^2}, \quad (1)$$

where M is the applied torque, h is the sample depth (23 mm), r and θ are the radius and the rotation angle in a cylindrical coordinate respectively. If we assume a non-Newtonian fluid with a power-law type flow relation,

$$\dot{\epsilon}_{r\theta} = k\tau_{r\theta}^n, \quad (2)$$

where $\dot{\epsilon}_{r\theta}$ is the strain rate and k is a constant, the rotation rate $\dot{\theta}$ in a steady state can be written as

$$\dot{\theta} = \frac{k}{2n} \left(\frac{M}{2\pi h} \right)^n \left(\frac{1}{R_i^{2n}} - \frac{1}{R_o^{2n}} \right), \quad (3)$$

where R_i and R_o are the radii of the rotation rod and the sample holder respectively [Watanebe *et al.*, 1992]. No slips were noted in the present study since there appeared no stress drops in the creep curves. The relation between the strain rate at R_i and the rotation rate of the rod is given by

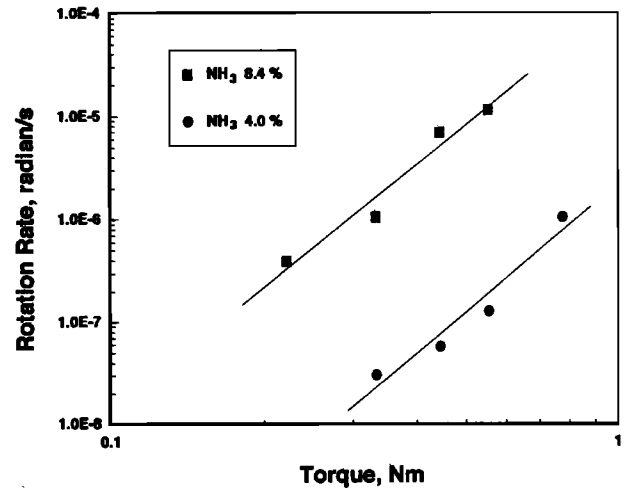


Figure 3. Minimum rotation rate versus applied torque. NH_3 contents are 4.0 and 8.4 % and temperature is 180 K. Fitted lines give $n = 4.0 \pm 0.1$.

$$\dot{\epsilon}_{r\theta}|_{R_i} = \frac{2nR_o^{2n}}{R_o^{2n} - R_i^{2n}} \dot{\theta}. \quad (4)$$

Results

Typical creep curves of partially melted ice are shown in Fig. 2, which clearly shows the transient, secondary and tertiary creeps as often noted in the ice rheology [Hobbs, 1974]. Our focus is put on the the minimum rotation rate appearing in the secondary creep region where a steady deformation proceeds. Fig. 2 shows that the steady-state rotation rates become larger as the temperature rises; the meaningful measurement becomes difficult at temperatures above 220 K because of slips between the rotation rod and the sample.

Fig. 3 shows the torque dependence of the minimum rotation rate for samples of NH_3 contents, 4.0 and 8.4 %, at 180 K. The dependence is in good agreement with eq.3 and the calculated value of n is 4.0 ± 0.1 ; the shear stress generated in the samples is estimated from 0.1 MPa to 10 kPa at an applied torque of 0.33 Nm. Kargel *et al.* [1991] shows that the value of n of partially melted ammonia-water mixture is from 1.0 to 2.4 when its melt fraction is higher than 80 %. In the sub-solidus condition, Durham *et al.* [1993] find $n = 5.6$ for a 4.7 % NH_3 sample. The result that our value is in between is reasonable because the melt fractions in our samples are roughly estimated as 20 %, much smaller than those of Kargel *et al.* [1991]. According to the deformation map of ice constructed by Goodman *et al.* [1977], the main deformation mechanism in the temperature and stress region of our study is motions of dislocations and volume diffusion. If the deformation of the partially melted ice of the ammonia-water system were controlled only by that of ice, the stress exponent should be smaller than 4, possibly in the range 1 - 3. Our result suggests that the contribution of a liquid phase between ice grains is important. However, the texture of partially melted ice could not be observed in our device. The microscopic observation of the sample texture is under way to make clear the effect of the liquid phase distribution on the viscosity properties.

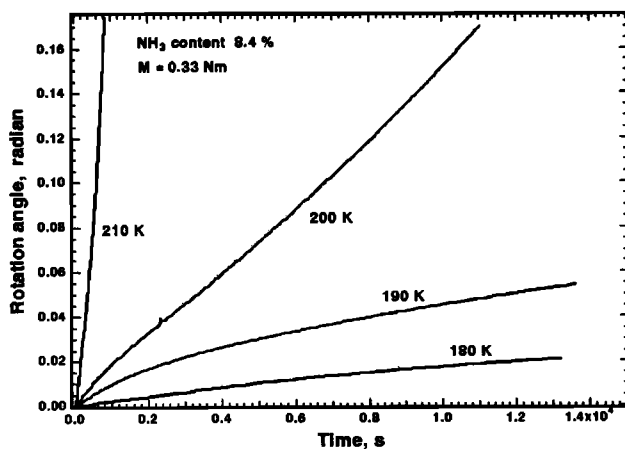


Figure 2. Rotation angle versus time. The applied torque is 0.33 Nm and NH_3 content is 8.4 %.

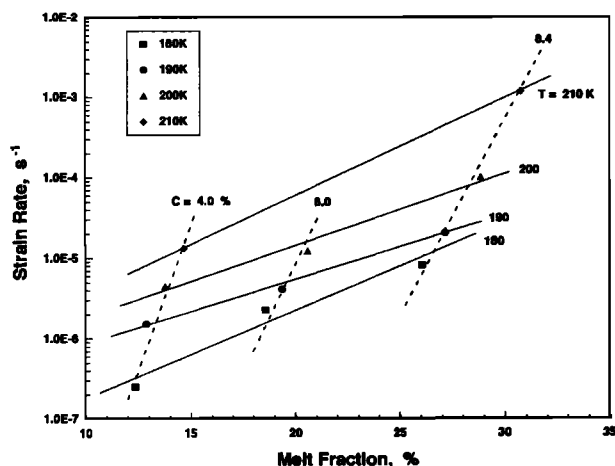


Figure 4. Minimum strain rate versus melt fraction. The applied torque is 0.33 Nm. The relation, $\dot{\epsilon} = a e^{b\phi}$, is described by solid and dashed lines for constant temperature (T) and NH_3 content (C) respectively.

In Fig. 4, the minimum strain rate is plotted as a function of the melt fraction and temperature. The minimum strain rate increases systematically with the increase of the melt fraction. As the melt fraction increases with NH_3 content and temperature, the dependence of the strain rate on the melt fraction is simply described by a relation, $\dot{\epsilon} = a e^{b\phi}$, where ϕ is the melt fraction, and a and b are constants. The fitted straight lines in the figure are those relations at constant temperatures (solid lines) and constant NH_3 contents (dashed lines).

Discussion

In the shear creep tests of the ammonia-water system, the minimum rotation rate, or the steady-state strain rate of partially melted ice was investigated in

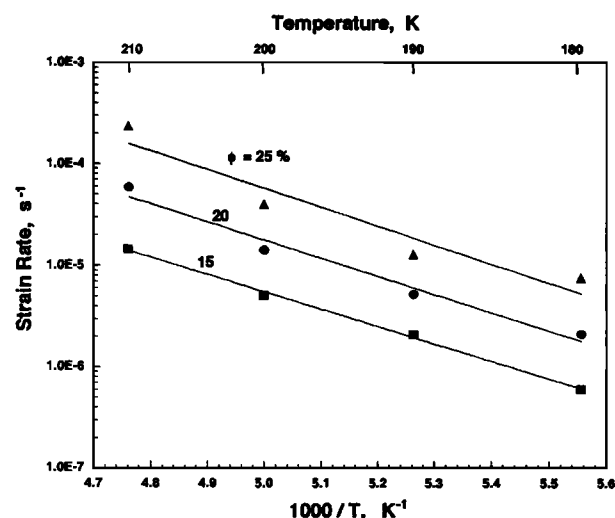


Figure 5. Minimum strain rate versus reciprocal of temperature. The melt fractions are $\phi = 15, 20$ and 25% . The activation energy estimated from the slopes of lines is 33.7 ± 0.8 kJ/mol.

a range of melt fraction, 12 - 30 %, at temperatures from 180 to 210 K. The stress exponent of the flow law obtained (eq.2) was 4.0 ± 0.1 , implying that other deformation mechanisms than usual diffusion and dislocation creep of ice might be responsible [Hobbs, 1974]. In Fig. 5, strain rates estimated at melt fractions, 15, 20 and 25 % in Fig. 4, are plotted against the reciprocal of temperature. The temperature variations thus estimated do not involve effects caused by the change in the melt fraction. If the deformation is controlled by a thermal activation process, the strain rate is written by the following Arrhenius equation,

$$\dot{\epsilon} = A \exp\left(-\frac{E}{RT}\right), \quad (5)$$

where R is the gas constant, E is the activation energy and A is a constant. The energy is calculated from slopes of the lines at each melt fraction, which is 33.7 ± 0.8 kJ/mol. This value is rather smaller than that of pure water ice, 70 kJ/mol (213 K~263 K, Arakawa and Maeno [1992]), and suggests the contribution of a liquid phase appearing between composing ice grains.

The activation energy of viscosity of NH_3 solution was estimated by Kargel *et al.* [1991], as 35 kJ/mol at temperatures below 240 K (NH_3 contents 21 - 34 %). Durham *et al.* [1993] obtained the activation energy in a sub-solidus condition, which was 54.1 kJ/mol for NH_3 content 4.7 % and temperature 143 - 176 K. The similarity of our activation energy to that of Kargel *et al.* [1991] suggests that a similar mechanism is contributing to the deformation process, that is the grain displacement or sliding associated with liquids. The liquid between ice grains, existing as films, conduits and pockets, may help grains to slip and arrange the structure when stress is applied. The non-linear flow law is considered to result from the rearrangement and interaction of grains.

Finally, the effective viscosity of the ammonia-water system (4 - 12 %) was estimated by the relation $\mu_{eff} = \tau_{r\theta} / \dot{\epsilon}_{r\theta}$, and shown in Fig. 6, which includes data by Kargel *et al.* [1991] and Durham *et al.* [1993]. Their triaxial compression viscosity (λ_{eff}) was transformed to the shear viscosity ($\mu_{eff} = \lambda_{eff}/3$) and the effective viscosity was estimated at a stress 0.1 MPa. The viscosity at temperatures below the peritectic point is large, $10^{17} \sim 10^{22}$ Pa s, and by 10~11 orders of magnitudes larger than ours with almost similar NH_3 contents, which is $10^7 \sim 10^{11}$ Pa s above the peritectic temperature. Above 210 K where melt fractions are large ($\geq 75\%$), the viscosity is very small, 0.2~1 Pa s. It is concluded that the viscosity of the ammonia-water mixture shows drastic changes at two temperatures, namely the peritectic point and between 210 K and 240 K. The latter temperature corresponds to a melt fraction larger than about 35 %.

The large change in viscosity of forming materials with temperature and composition might cause important effects on the thermal histories of many icy satellites. The graben system on Ariel is a tectonic system formed by lithospheric extension and down-dropping along normal faults. The grabens are observed to be partially filled by the smooth, lobate flows [Croft and

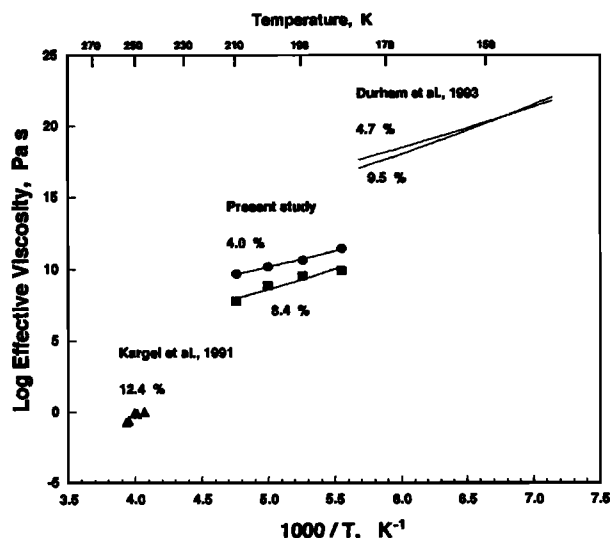


Figure 6. Effective viscosity of ammonia-water mixture versus reciprocal of temperature. The NH_3 content is around 10 %; 4.7–9.5 %, Durham *et al.* [1993]; 4.0–8.4 %, the present study; 12.4 %, Kargel *et al.* [1991]. The effective viscosity was estimated at 0.1 MPa. A Newtonian fluid was assumed for data of Kargel *et al.*

Soderblom, 1989]. According to analyses of Voyager's image data by Jankowski and Squyres [1988] and Melosh and Janes [1989], the estimated viscosity of the flows depends on the models assumed and ranged from about 10^{16} Pa s for a solid flow to 10^{13} – 10^{15} Pa s for a partially melted flow. If we assume that the main composition of the surface crust of Ariel is ammonia-water mixtures and that the bulk lava flow is rheologically homogeneous, our result means that the viscosity to form the grooved structures might be only explained by the partially melted ice with a lower melt fraction than 30 %. However, this is one possible explanation because the real lava flows are not homogeneous and have lava crusts which have important effects on the deformational behavior of bulk lava flows [Greenberg *et al.*, 1989].

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